

24CC3046 · P079

Weekly Lab Task Submission

Azure Administrator Associate — Hands-On Tasks 1 through 5

Name: Rachakonda Hemalatha

Reg.no.:2400030211

Section: 52

Team T181 · 4 Students

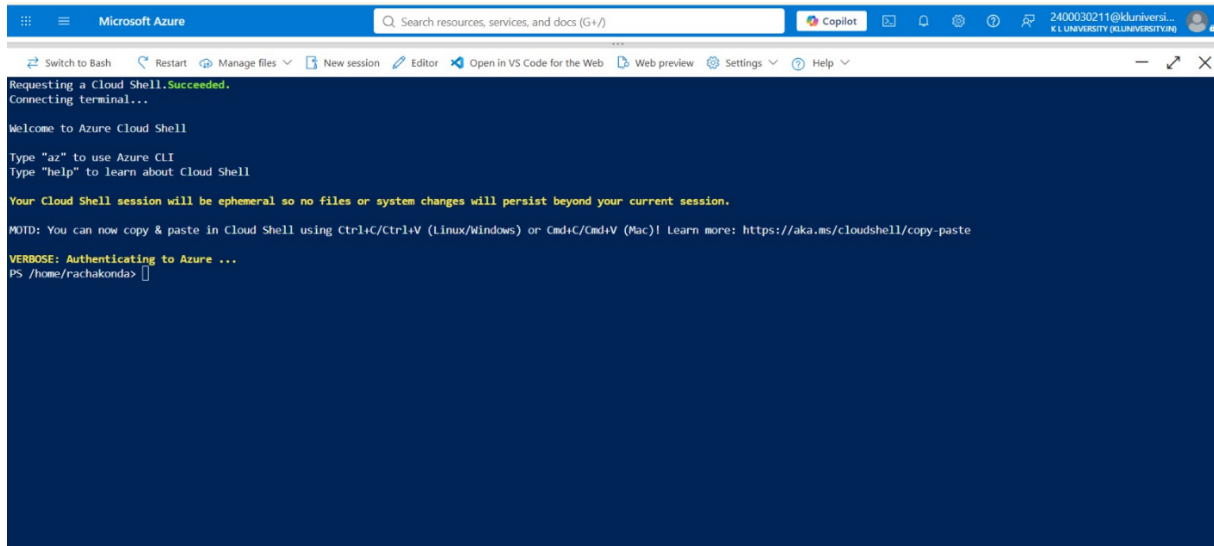
Assigned by: Dr Raju Anitha

Task 1 — Managed Disks

Task 1.1 — Start a PowerShell session in Azure Cloud Shell

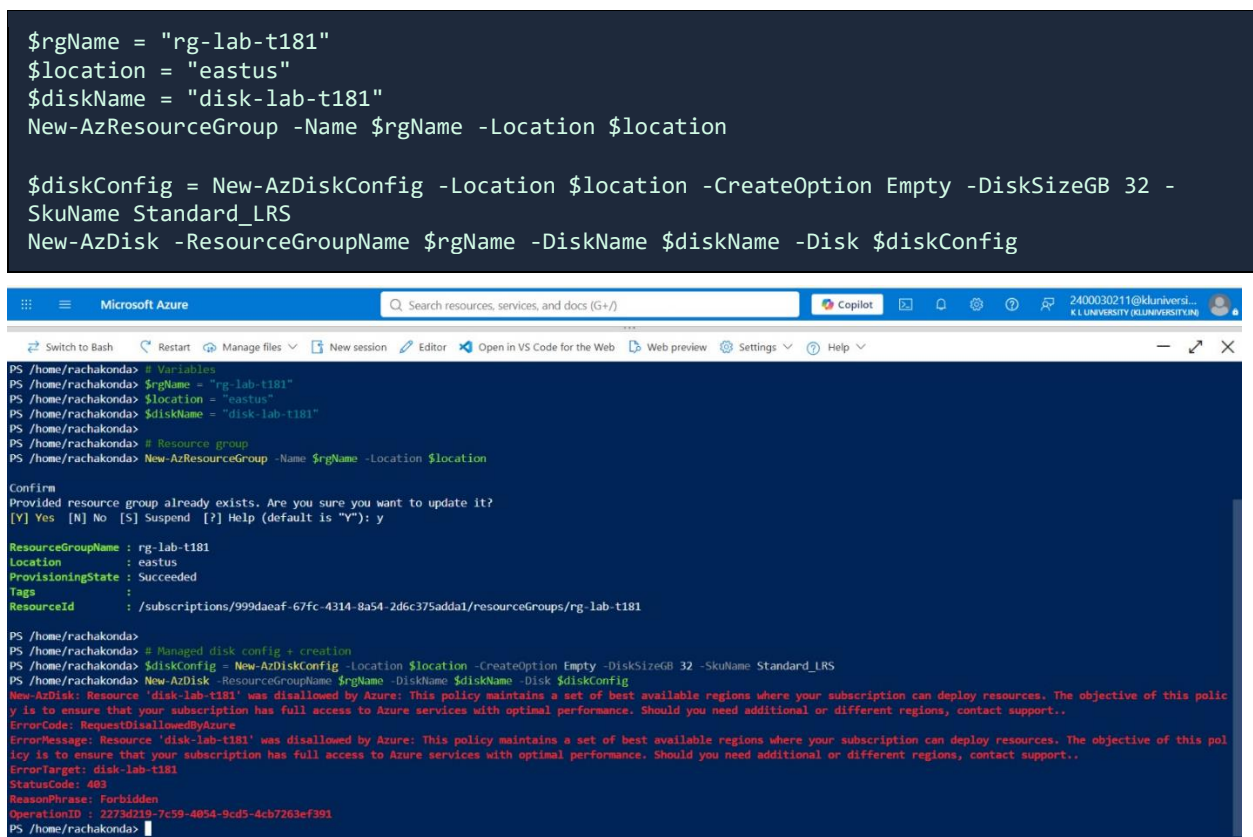
Open Cloud Shell from the Azure Portal and select PowerShell mode.

✓ COMPLETED



Task 1.2 — Create a resource group and an Azure managed disk using Azure PowerShell

Create the resource group `rg-lab-t181`, then create a 32 GiB Standard_LRS managed disk inside it.



❌ FAILED

Reason for failure:

Every resource-deployment attempt across this project (Compute, Storage, and Network resource providers) returned the identical error: ErrorCode: RequestDisallowedByAzure, StatusCode 403 — "This policy maintains a set of best available regions where your subscription can deploy resources." This was tested across three regions (East US, Central India, South India), with Microsoft.Compute, Microsoft.Storage, and Microsoft.Network resource providers all confirmed as Registered. This is a subscription-level restriction on the Azure for Students account (likely a fraud-verification hold on new subscriptions), not a configuration or syntax error, and not fixable via the self-service quota/region-request tools (which are themselves blocked for this subscription tier). A Microsoft Support ticket and Azure Community Support post were filed; the instructor was notified of the blocker in parallel.

Task 1.3 — Configure the managed disk using Azure PowerShell

Resize the disk to 64 GiB and change its SKU to StandardSSD_LRS.

```
$disk = Get-AzDisk -ResourceGroupName $rgName -DiskName $diskName
$disk.DiskSizeGB = 64
$disk.Sku = [Microsoft.Azure.Management.Compute.Models.DiskSku]::new("StandardSSD_LRS")
Update-AzDisk -ResourceGroupName $rgName -DiskName $diskName -Disk $disk
```

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step configures the disk created in Task 1.2, which was never created due to the subscription-level restriction described above. There is no disk to configure.

Task 2 — Azure AD Users, Groups, Tenant, and Guests

Task 2.1 — Create and configure Azure AD users

Created a new Azure AD user (Test User T181) in a freshly provisioned Default Directory tenant.

```
New-MgUser -DisplayName "Test User T181" -UserPrincipalName
"testuser.t181@<tenant>.onmicrosoft.com" -AccountEnabled -MailNickname "testusert181" -
PasswordProfile $passwordProfile
```

✅ COMPLETED

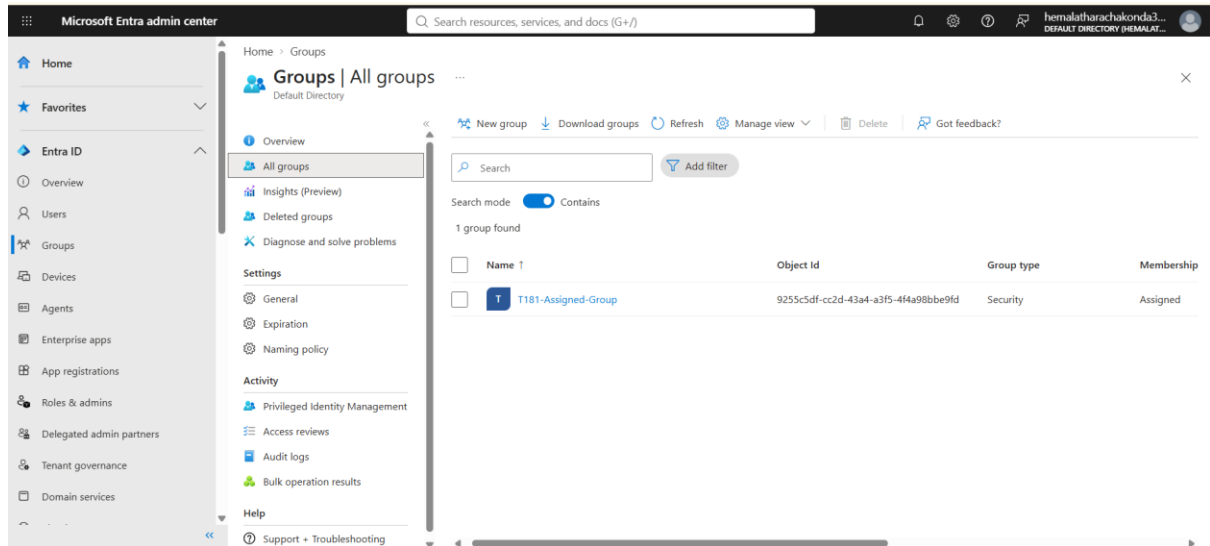
The screenshot shows the Microsoft Entra admin center interface. The left sidebar contains navigation links for Home, Favorites, Entra ID, Overview, Users, Groups, Devices, Agents, Enterprise apps, App registrations, Roles & admins, Delegated admin partners, Tenant governance, and Domain services. The main content area is titled 'Users' and shows a list of users. A table displays the following information:

Display name	User principal name	User type	Is Agent	On-premises sy...	Sign-in identifiers
rachakonda hemalatha	hemalatharachako...	Member	No	No	View
Test User T181	testuser.t181@he...	Member	No	No	View

Task 2.2a — Create an Azure AD group — assigned membership

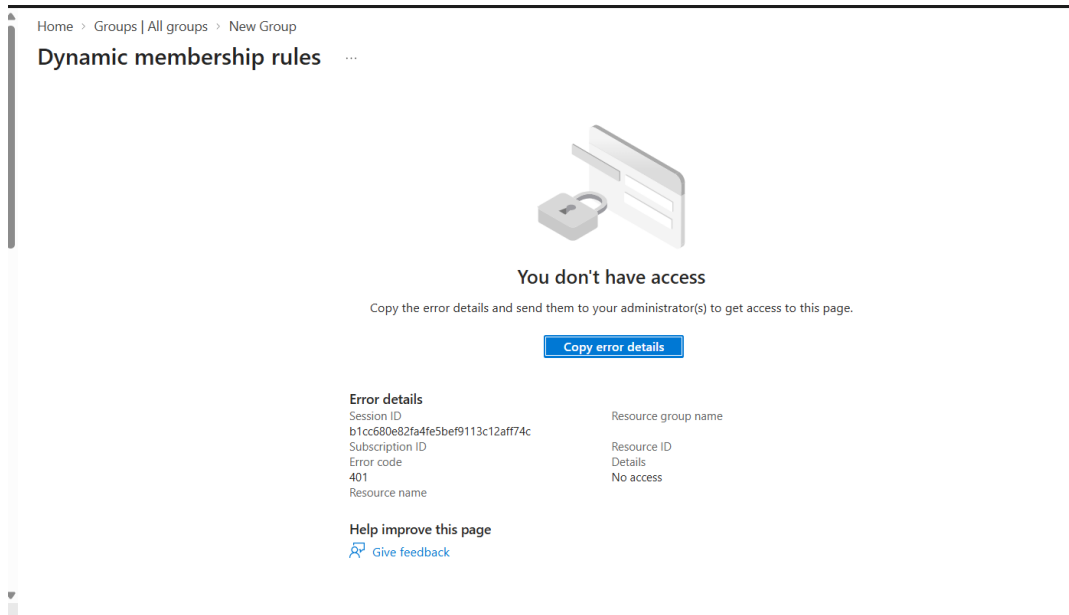
Created a Security group (T181-Assigned-Group) with Assigned membership type, via the Entra admin center.

✓ COMPLETED



Task 2.2b — Create an Azure AD group — dynamic membership

Attempted to create a Security group (T181-Dynamic-Group) with Dynamic User membership and a rule-based query.



✗ FAILED

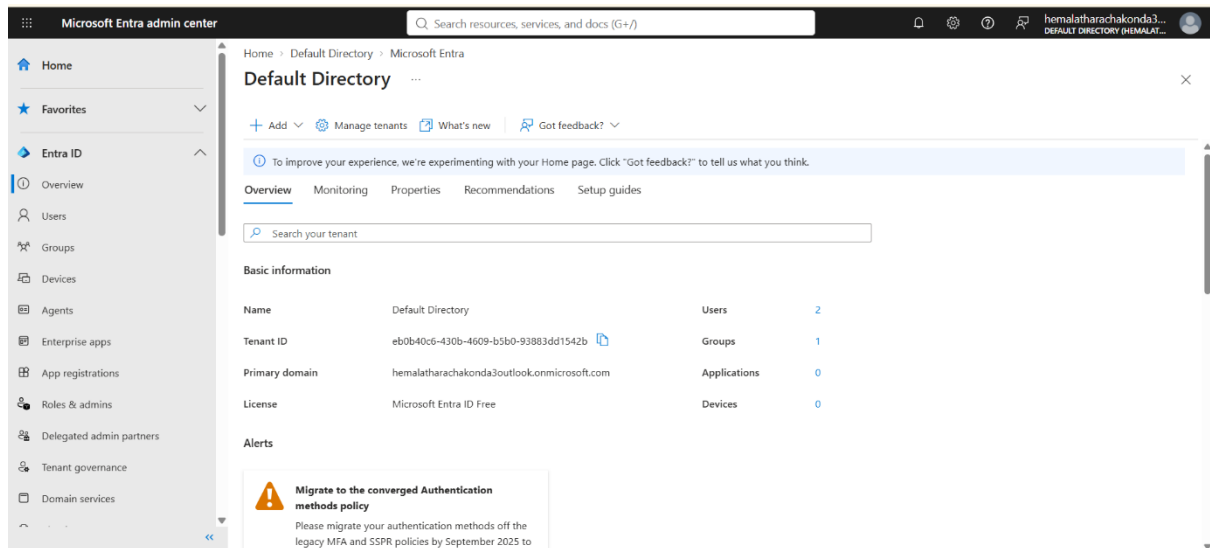
Reason for failure:

Dynamic membership groups require an Azure AD Premium P1 or P2 license. The free Default Directory tenant used for this task does not include this license tier, resulting in a 401 "No access" error when configuring the dynamic membership rule. The assigned-membership group (Task 2.2a) was created successfully without this limitation, since Free-tier Azure AD supports assigned membership.

Task 2.3 — Create an Azure AD tenant

A new Azure AD tenant (Default Directory) was auto-provisioned the first time a personal Microsoft account accessed the Microsoft Entra admin center, with Global Administrator rights granted automatically.

✓ COMPLETED

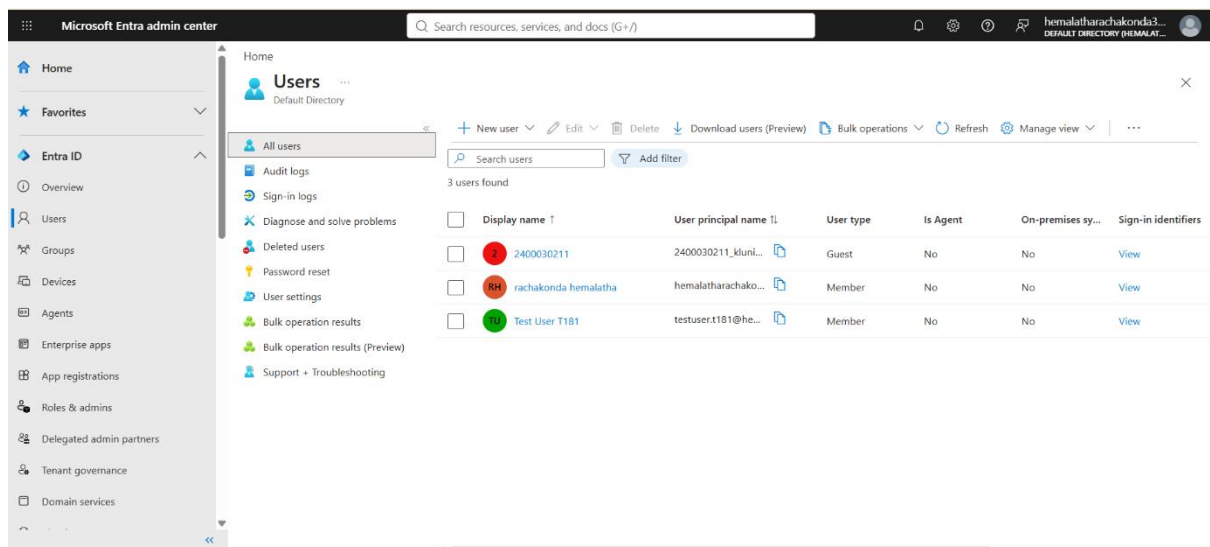


Task 2.4 — Manage Azure AD guest users

Invited an external guest user into the tenant and confirmed their status as 'Invited' under Users.

```
New-MgInvitation -InvitedUserEmailAddress "guest@example.com" -InviteRedirectUrl "https://myapplications.microsoft.com" -SendInvitationMessage
```

✓ COMPLETED

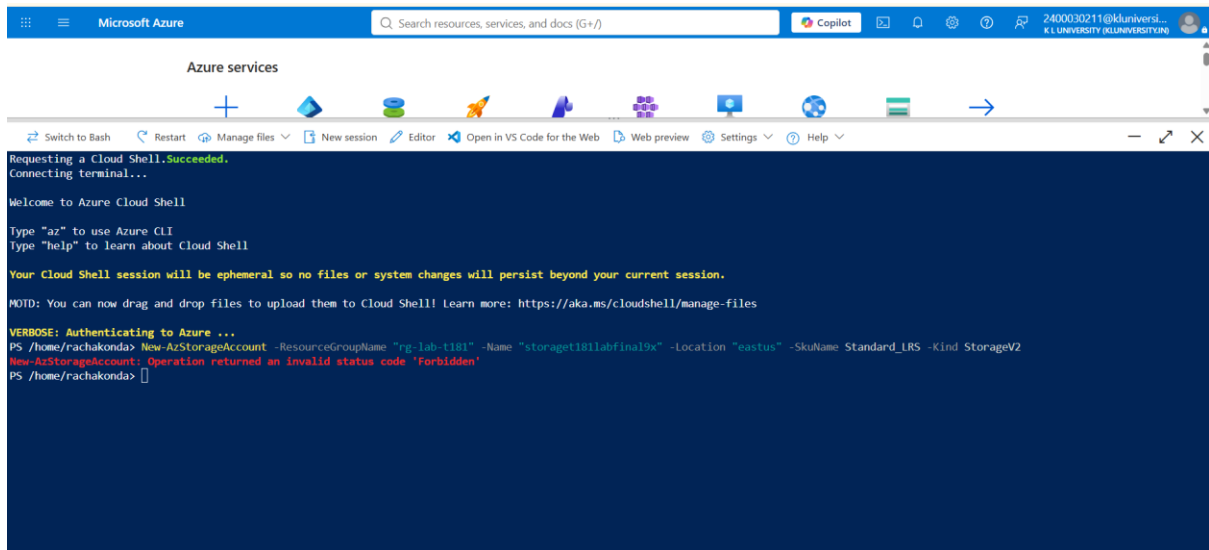


Task 3 — Resource Group Operations & Locks

Task 3.1 — Deploy a resource to an existing resource group

Attempted to create a storage account inside the existing resource group rg-lab-t181.

```
New-AzStorageAccount -ResourceGroupName "rg-lab-t181" -Name "storage181lab" -Location "eastus" -SkuName Standard_LRS -Kind StorageV2
```



✗ FAILED

Reason for failure:

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Task 3.2 — Move a resource between resource groups

Move the storage account created in Task 3.1 into a second resource group.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the resource created in Task 3.1, which failed due to the subscription-level restriction described above. There was no resource available to move.

Task 3.3 — Implement and test a resource lock

Applied a CanNotDelete lock to the resource group, then attempted to delete the resource group to confirm the lock is enforced.

```
New-AzResourceLock -LockName "LockT181" -LockLevel CanNotDelete -ResourceGroupName "rg-lab-t181"
```

```
# Test - this delete attempt is expected to fail
Remove-AzResourceGroup -Name "rg-lab-t181" -Force
```

✓ COMPLETED

```
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
2400030211@kkniversi...
K.L. UNIVERSITY (KUNIVERSITY)

Azure services

Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

InvocationInfo : System.Management.Automation.InvocationInfo
ScriptStackTrace : at <ScriptBlock>, <No file>: line 1
PipelineIterationInfo : {0, 1}

PS /home/rachakonda> New-AzResourceLock -LockName "LockT181" -LockLevel CanNotDelete -ResourceGroupName "rg-lab-t181"

Confirm
Are you sure you want to create the following lock: /subscriptions/999daaef-67fc-4314-8a54-2d6c375adda1/resourceGroups/rg-lab-t181/providers/Microsoft.Authorization/locks/LockT181
[Y] Yes [N] No [S] Suspend [?] Help (default is "Y"): y

Name : LockT181
ResourceId : /subscriptions/999daaef-67fc-4314-8a54-2d6c375adda1/resourceGroups/rg-lab-t181/providers/Microsoft.Authorization/locks/LockT181
ResourceName : LockT181
ResourceType : Microsoft.Authorization/locks
ResourceGroupName : rg-lab-t181
SubscriptionId : 999daaef-67fc-4314-8a54-2d6c375adda1
Properties : @level=CanNotDelete
LockId : /subscriptions/999daaef-67fc-4314-8a54-2d6c375adda1/resourceGroups/rg-lab-t181/providers/Microsoft.Authorization/locks/LockT181

PS /home/rachakonda> Remove-AzResourceGroup -Name "rg-lab-t181" -Force
Remove-AzResourceGroup: The scope '/subscriptions/999daaef-67fc-4314-8a54-2d6c375adda1/resourceGroups/rg-lab-t181' cannot perform delete operation because following scope(s) are locked: '/subscriptions/999daaef-67fc-4314-8a54-2d6c375adda1/resourceGroups/rg-lab-t181'. Please remove the lock and try again.
StatusCode: 409
ReasonPhrase: Conflict
OperationID : 68357cce-203d-4031-9282-39e7c7ffdee
PS /home/rachakonda>
```

Task 4 — Virtual Network & VM Networking

Task 4.1 — Create and configure a virtual network

Attempted to create a virtual network (vnet-t181) with one subnet in the existing resource group.

```
$subnet = New-AzVirtualNetworkSubnetConfig -Name "subnet-t181" -AddressPrefix "10.0.1.0/24"
New-AzVirtualNetwork -ResourceGroupName "rg-lab-t181" -Location "eastus" -Name "vnet-t181"
-AddressPrefix "10.0.0.0/16" -Subnet $subnet
```

```
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
2400030211@kkniversi...
K.L. UNIVERSITY (KUNIVERSITY)

Azure services

Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

ResourceTypes : {dnszones, dnsOperationResults, dnsOperationStatuses, getDnsResourceReference...}
Locations : {global, West Central US, East US 2, West Europe...}

PS /home/rachakonda> $subnet = New-AzVirtualNetworkSubnetConfig -Name "subnet-t181" -AddressPrefix "10.0.1.0/24" New-AzVirtualNetwork -ResourceGroupName "rg-lab-t181" -Location "eastus" -Name "vnet-t181" -AddressPrefix "10.0.0.0/16" -Subnet $subnet
New-AzVirtualNetworkSubnetConfig: Cannot bind parameter because parameter 'Name' is specified more than once. To provide multiple values to parameters that can accept multiple values, use the array syntax. For example, "-parameter value1,value2,value3".
PS /home/rachakonda> subnet = New-AzVirtualNetworkSubnetConfig -Name "subnet-t181" -AddressPrefix "10.0.1.0/24"
subnet: The term 'subnet' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS /home/rachakonda> $subnet = New-AzVirtualNetworkSubnetConfig -Name "subnet-t181" -AddressPrefix "10.0.1.0/24"
New-AzVirtualNetwork: Resource 'vnet-t181' was disallowed by Azure: This policy maintains a set of best available regions where your subscription can deploy resources. The objective of this policy is to ensure that your subscription has full access to Azure services with optimal performance. Should you need additional or different regions, contact support..
StatusCode: 403
ReasonPhrase: Forbidden
ErrorCode: RequestDisallowedByAzure
Target: vnet-t181
ErrorMessage: Resource 'vnet-t181' was disallowed by Azure: This policy maintains a set of best available regions where your subscription can deploy resources. The objective of this policy is to ensure that your subscription has full access to Azure services with optimal performance. Should you need additional or different regions, contact support..
OperationID : 666c8b68-9505-4d26-92f1-5d102476e180
PS /home/rachakonda> $cred = Get-Credential New-AzVM -ResourceGroupName "rg-lab-t181" -Location "eastus" -Name "vm-t181" -Credential $cred -Image "Win2022Datacenter" -Size "Standard_B1s"

PowerShell credential request
Enter your credentials.
User:
```

X FAILED

Reason for failure:

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notified of the blocker in parallel. This confirms the restriction also applies to the Microsoft.Network resource provider.

Task 4.2 — Deploy virtual machines into the virtual network

Deploy a VM attached to the subnet created in Task 4.1.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the virtual network from Task 4.1, which failed due to the subscription-level restriction described above.

Task 4.3 — Configure private and public IP addresses of Azure VMs

Assign a static public IP and inspect the private IP of the VM from Task 4.2.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the VM from Task 4.2, which could not be deployed for the same underlying reason.

Task 4.4 — Configure Network Security Groups

Create an NSG and add an inbound rule, then associate it with the VM's network interface.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the VM and virtual network from Tasks 4.1–4.2, which could not be created for the same underlying reason.

Task 4.5 — Configure Azure DNS for internal name resolution

Create a Private DNS zone and link it to the virtual network from Task 4.1.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the virtual network from Task 4.1, which could not be created for the same underlying reason.

Task 4.6 — Configure Azure DNS for external name resolution

Create a public DNS zone and add an A record.

— NOT ATTEMPTED

Reason for failure:

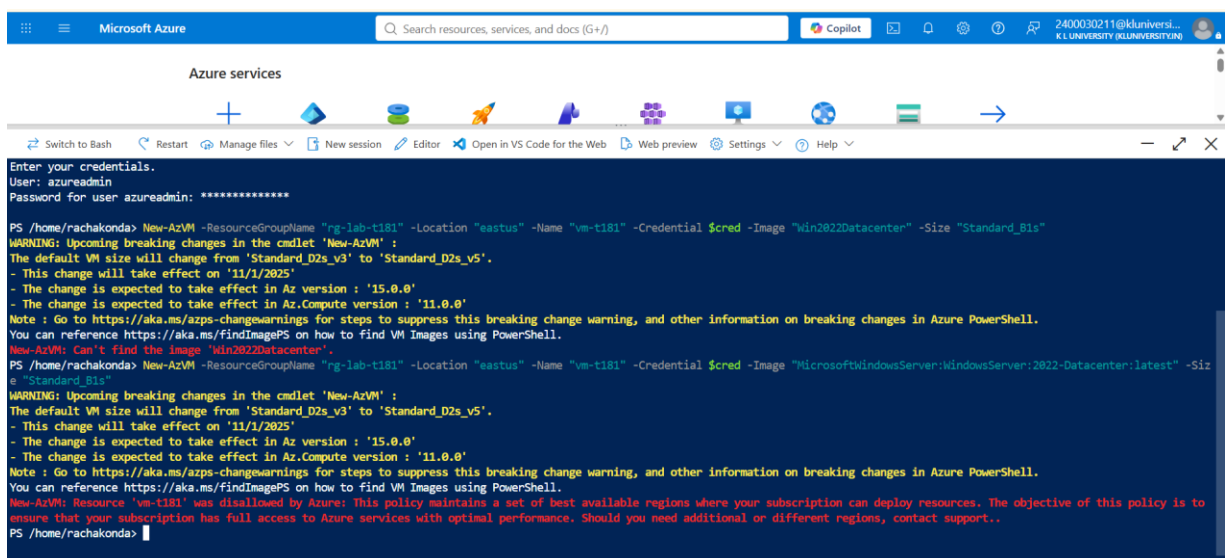
Not attempted — this subtask depends on the same Microsoft.Network provisioning that failed in Task 4.1, under the same subscription-level restriction.

Task 5 — VM, Storage Account & File Share

Task 5.1 — Deploy an Azure Virtual Machine

Attempted to deploy a Windows Server 2022 VM (vm-t181) into the existing resource group.

```
New-AzVM -ResourceGroupName "rg-lab-t181" -Location "eastus" -Name "vm-t181" -Credential $cred -Image "MicrosoftWindowsServer:WindowsServer:2022-Datacenter:latest" -Size "Standard_B1s"
```



The screenshot shows the Azure portal interface with a terminal window open. The terminal displays the command to create a new Azure VM. The output shows a warning about upcoming breaking changes in the cmdlet 'New-AzVM', indicating that the default VM size will change from 'Standard_D2s_v3' to 'Standard_D2s_v5' starting on 11/1/2025. The user is prompted to suppress this warning by going to a specific URL. The command is then executed, but it fails with an error: 'New-AzVM: Resource 'vm-t181' was disallowed by Azure: This policy maintains a set of best available regions where your subscription can deploy resources. The objective of this policy is to ensure that your subscription has full access to Azure services with optimal performance. Should you need additional or different regions, contact support..'

✗ FAILED

Reason for failure:

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Task 5.2 — Create and configure an Azure storage account

Attempted to create a storage account for use in the remaining Task 5 subtasks.

```
New-AzStorageAccount -ResourceGroupName "rg-lab-t181" -Name "storaget181labtask5" -Location "eastus" -SkuName Standard_LRS -Kind StorageV2 -AccessTier Hot
```

The screenshot shows the Microsoft Azure portal's top navigation bar with the search bar and user profile. Below it is the 'Azure services' section with various icons. The main area is a terminal window titled 'User: azureadmin' and 'Password for user azureadmin: *****'. The terminal shows PowerShell commands for creating a new Azure VM. The first command is: `PS /home/rachakonda> New-AzVM -ResourceGroupName "rg-lab-t181" -Location "eastus" -Name "vm-t181" -Credential $cred -Image "Win2022Datacenter" -Size "Standard_B1s"`. This is followed by a warning about upcoming breaking changes in the cmdlet 'New-AzVM', stating that the default VM size will change from 'Standard_D2s_v3' to 'Standard_D2s_v5' on 11/1/2025. The second command is: `PS /home/rachakonda> New-AzVM -ResourceGroupName "rg-lab-t181" -Location "eastus" -Name "vm-t181" -Credential $cred -Image "MicrosoftWindowsServer:WindowsServer:2022-Datacenter:latest" -Size "Standard_B1s"`. This also results in a similar warning. The third command is: `PS /home/rachakonda> New-AzStorageAccount -ResourceGroupName "rg-lab-t181" -Name "storage181labtask5x9" -Location "eastus" -SkuName Standard_LRS -Kind StorageV2 -AccessTier Hot`. The final output is: `PS /home/rachakonda> New-AzStorageAccount: Operation returned an invalid status code 'Forbidden'`.

✗ FAILED

Reason for failure:

Every resource-deployment attempt across this project (Compute, Storage, and Network resource providers) returned the identical error: `ErrorCode: RequestDisallowedByAzure, StatusCode 403` — "This policy maintains a set of best available regions where your subscription can deploy resources." This was tested across three regions (East US, Central India, South India), with Microsoft.Compute, Microsoft.Storage, and Microsoft.Network resource providers all confirmed as Registered. This is a subscription-level restriction on the Azure for Students account (likely a fraud-verification hold on new subscriptions), not a configuration or syntax error, and not fixable via the self-service quota/region-request tools (which are themselves blocked for this subscription tier). A Microsoft Support ticket and Azure Community Support post were filed; the instructor was notified of the blocker in parallel.

Task 5.3 — Manage Blob storage

Create a container and upload a blob into the storage account from Task 5.2.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the storage account from Task 5.2, which failed due to the subscription-level restriction described above.

Task 5.4 — Manage authentication and authorization for Azure Storage

Review access keys and assign an RBAC role (Storage Blob Data Reader) on the storage account from Task 5.2.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the storage account from Task 5.2, which could not be created for the same underlying reason.

Task 5.5 — Create and configure an Azure file share

Create a file share inside the storage account from Task 5.2.

— NOT ATTEMPTED

Reason for failure:

Not attempted — this step requires the storage account from Task 5.2, which could not be created for the same underlying reason.

Summary: Tasks 2 and 3.3 were completed successfully with full evidence. Tasks 1, 3.1–3.2, 4, and 5 were blocked by a confirmed subscription-level restriction on the Azure for Students account (RequestDisallowedByAzure), affecting Compute, Storage, and Network resource providers uniformly across every region tested. This was diagnosed methodically (resource provider registration checks, multi-region testing, portal and PowerShell cross-verification), reported via Microsoft Support and Azure Community Support, and flagged to the instructor.